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M15 Project Dossier — RAG for the Spending Analyzer

Project: Spending Analyzer, a tool that sorts a college student's bank charges into spending categories and helps them budget.

Meeting 15 question: Does retrieving evidence improve our project's answers, or only make them look more credible?

Files: M15_RAG_experiment.ipynb (all numbers below), data/transactions.csv, RAG_Handout.docx (worksheet).

1. What we learned in Meeting 15

RAG (retrieval-augmented generation) works like an open-book exam. Before answering, the system looks up the few passages from its sources that best match the question, then writes an answer using only those passages and says which passage each claim came from. The point is that every answer can be checked against its source.

We tried this on the question our A2 work left open: can a student's own past labels settle stores that sell more than one kind of thing, like Target? What we took away:

- **Retrieval helps when the evidence is there.** Pulling several past Target charges let the system see that small trips were food and large trips were dorm supplies, and pick correctly.
- **A citation is not proof.** An answer can cite a real chunk that says nothing about the question. We had to read each cited chunk to tell a supported answer from one that only looked supported.
- **Small steps before retrieval matter.** Our text cleanup deleted any word containing a digit, which erased a seller's name and led to a confident answer built on the wrong evidence.
- **How you cut the sources matters.** Guide keywords stored one per chunk were easy to find. When all of a category's keywords were stored as one chunk, the right keyword got buried and the match score fell below our cutoff.
- **Knowing when to stop is part of the job.** A minimum match score and an "ask the student" option kept several bad answers from going out.

2. Experiment contract

What we agreed to test, fixed before we ran anything:

Item	What we used
Project function	When a new charge comes in, suggest a category and point to the past charge or guide entry it's based on. If the evidence isn't good enough, ask the student.
Sources	S1 — Charge history: the student's 143 labeled charges from July–August 2026, one chunk per charge, with amount, category and note. IDs H-###. S2 — Category guide: our A2 keyword list, one keyword per chunk (112 chunks). IDs G-CATEGORY-##.
Test set	The 68 charges from September 2026, kept out of the sources so the system had never seen them. Five were picked for the main table, one per case type.
Pipeline	Bank text → cleanup (same as A2) → similarity search on 3–5 letter pieces of text (TF-IDF, cosine score 0 to 1) → top-k chunks → answer step → category + chunk IDs.
Answer step	Ignore matches under 0.50. If the remaining chunks agree, answer and cite them. If they disagree, use the past charge closest in amount (within 50%). Otherwise ask the student.
A/B variable	Top-k only. Condition A = 1 chunk. Condition B = 5 chunks.
Held fixed	Sources, chunking, cleanup, similarity method, 0.50 cutoff, answer rule, test charges, and how we judged results.
Success	Right category, with a cited chunk from the same store and the same kind of purchase.
Unsafe failure	A confident answer whose citation is about a different store or purchase, especially when no real evidence exists.

Item	What we used
Simulated	The answer step is a fixed rule standing in for an LLM, so results are repeatable and citations can be checked automatically. Retrieval and data are real (our data is made up, as in A2).

3. Evidence

Five test cases

Case	Expected evidence	A (top-k = 1)	B (top-k = 5)	Verdict
1 · Direct Price Chopper $74.32 (Groceries)	Past Price Chopper trips	H-009 → Groceries. Right.	4 Price Chopper chunks + guide → Groceries. Right.	Both supported. No failure.
2 · Paraphrase Hudson Valley Urgent Care $35.00 (Health)	G-HEALTH-04 (URGENT CARE)	G-HEALTH-04 (0.77) → Health. Right.	Same, plus 4 paycheck chunks that share "HUDSON" (0.24, cut). Right.	Both supported. Unrelated chunks pulled in but filtered (R3).
3 · Multi-part Target $61.44 (Shopping, dorm stuff)	H-049, H-126 ($63 dorm trips)	H-011, a $20 food run (tie, oldest won) → Groceries. Wrong.	All 4 Target chunks; closest amount H-049 → Shopping. Right.	A missed the evidence (R1). B supported.
4 · Conflicting CVS $13.93 (Health, cold medicine)	H-037 (allergy meds)	H-037 → Health. Right, by tie order.	Health $22.10 vs Shopping $9.48; went by amount → Shopping. Wrong.	B used the wrong signal (R3 + G1).
5 · Out-of-scope PayPal *KJ8STUDIO $18.00 (Shopping, unknown seller)	None exists	H-079, an eBay purchase (0.59) → Shopping	Same H-079 → Shopping	Both unsupported: cites a different seller (R3 + C1).

Counts across the five cases

Measure	A (top-k = 1)	B (top-k = 5)
Relevant evidence reached the answer step	3 / 5	4 / 5
Correct category	4 / 5	4 / 5
Claim supported by its citation	3 / 5	3 / 5
Said "not enough evidence" on case 5	No	No
All 68 September charges	64 right, 4 wrong	65 right, 2 wrong, 1 sent to student

Claim-to-chunk checks

- **Supported:** the answer claims Target $61.44 is Shopping; chunk H-049 says "TARGET LATHAM NY, $63.13, Shopping, apartment / dorm stuff"; so the claim is supported.
- **Unsupported:** the answer claims PayPal *KJ8STUDIO is Shopping; chunk H-079 says "PAYPAL * EBAY INC, $31.50, Shopping"; that is a different seller, so the claim is unsupported, even though the category happens to be right.

Other things we checked

- **Full month:** going to 5 chunks fixed Target $61.44 and Stewart's $5.53, turned a wrong Apple $0.99 answer into "ask the student," and broke CVS. Costco $64.99 (a desk chair) was wrong in both, because all our Costco evidence says groceries.
- **Chunk size (extra test):** with one guide chunk per category instead of per keyword, Urgent Care dropped from 0.77 to 0.33 and Ticketmaster from 0.96 to 0.41, both under the cutoff. Uniqlo matched a New York subway charge instead.

4. Decision

Decision: MODIFY

We will modify RAG for suggesting categories for new charges because condition B (top-k = 5) retrieved both kinds of past Target trips and the answer cited the matching $63 dorm-stuff trip. The main risk is confident answers that cite a different store when we have no real evidence (the PayPal case), so next we will keep store names that contain digits during cleanup, only accept a citation from the same store, and send stores like CVS to the student.

Compared with simpler options

Option	What we think
Fixed rules, no retrieval	Fine for one-purpose stores and bills (Netflix, rent). Fails at mixed stores, and our A2 "FEE" rule even tagged the gym as Fees.
Search only	Showing the student the five closest past charges with their notes may be enough for stores like Target or CVS, since the student knows what they bought.
LLM only (A2)	Agreed with our labels on 93–94% of these same charges but couldn't show where its answers came from.
RAG	96% on the same charges, with a source for every answer. Not a fair race, though: RAG could read the student's own history and the LLM couldn't. Worth keeping for one-line suggestions if we block answers that cite a different store.

5. Limits & Next Question

Limits

- Our data is made up (211 charges for one student over three months), so the numbers show how the method behaves, not how it would do on real statements.
- The answer step was a fixed rule, not an LLM. A real LLM might handle the conflicting cases better or worse.
- Five cases is a small sample. The full-month run helps, but September only has a handful of mixed-store charges.
- The similarity search matches letters, not meaning, and ties are broken by whichever chunk was saved first.
- Every past label was treated as correct. A real app's saved labels could include mistakes or unchecked guesses.

Next question

If we require each citation to come from the same store, and send mixed stores where price doesn't separate purchases to the student, do we keep RAG's gains without confident answers built on the wrong evidence?

Next tests

- Fix the cleanup step so store names with digits (like KJ8STUDIO) survive, then rerun the five cases and the full month.
- Add the same-store check on citations and count how many answers turn into "ask the student."
- Run a real LLM as the answer step on the same five cases, using the prompt the notebook prints, and compare its citations to the rule's.